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Quantum gravimeters for hydrocarbon reservoir analysis

Abstract

The disclosure presents processes to locate one or more quantum gravimeters at a hydrocarbon well site, with at least one quantum gravimeter at a surface location. One or more additional gravimeters, whether quantum gravimeters or non-quantum gravimeters, can be located downhole a wellbore of the well site. Gravitational data collected from various gravimeters can be analyzed to produce analyzed gravitational parameters and subterranean formation parameters. In some aspects, the gravitational data can be processed, such as by an inversion algorithm or a noise reduction algorithm. The generated results can be used to calibrate non-quantum gravimeters located proximate the well site or downhole the wellbore, identify a depth and direction of a water front, identify the fluid flow of hydrocarbons or water in the subterranean formation, to identify orphaned hydrocarbon reservoirs, or other characteristics of fluid flow or subterranean formation parameters, such as subterranean formation damage.

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Background/Summary

TECHNICAL FIELD

(1) This application is directed, in general, to analyzing hydrocarbon reservoirs and, more specifically, to employing quantum gravimeters to aid analysis.

BACKGROUND

(2) In developing a hydrocarbon reservoir or performing operations for hydrocarbon production, various types of fluids are often pumped downhole and fluids, hydrocarbons, and subterranean formation material can be removed from the downhole location. These changes in the mass at various locations within the subterranean formation can be measured and used as inputs to future stages of developing or producing hydrocarbons within the reservoir. Conventional gravimeters can be used to measure the change in gravity at various points of the subterranean formation. Conventional gravimeters are subject to drift and other issues that reduce their effectiveness for

well site operations. It would be beneficial to have a more accurate and efficient way to measure changes in gravity of a reservoir.

SUMMARY

(3) In one aspect, a method is disclosed. In one embodiment, the method includes (1) locating one or more surface quantum gravimeters at a surface location proximate a well site of a reservoir, where the well site is for moving fluids into or out of the reservoir and has one or more wellbores, (2) collecting gravitational data at the one or more surface quantum gravimeters, and (3) processing the gravitational data received from the one or more surface quantum gravimeters, generating analyzed gravitational parameters and subterranean formation parameters that indicate mass changes within a subterranean formation of the reservoir as a stage of a well site operation plan is executed.

(4) In a second aspect, a system is disclosed. In one embodiment, the system includes (1) a set of quantum gravimeters positioned to collect gravitational data of a subterranean formation of a reservoir, wherein the reservoir has well sites for hydrocarbon extraction, (2) a data transceiver, capable of receiving the gravitational data from the set of quantum gravimeters, and (3) a gravitational processor, capable of communicating with the data transceiver and analyzing the gravitational data to generate analyzed gravitational parameters and one or more subterranean formation parameters.

(5) In a third aspect, a computer program product having a series of operating instructions stored on a non-transitory computer-readable medium that directs a data processing apparatus when executed thereby to perform operations to determine subterranean formation parameters and analyzed gravitational parameters is disclosed. In one embodiment, the computer program product has operations that include (1) receiving collected gravitational data from one or more surface quantum gravimeters, wherein the one or more surface quantum gravimeters are located at a surface location proximate a well site of a reservoir, where the well site is for hydrocarbon production and has one or more wellbores, (2) processing the gravitational data received from the one or more surface quantum gravimeters, generating the analyzed gravitational parameters and the subterranean formation parameters that indicate fluid movement in a subterranean formation of the reservoir as a stage of a well site operation plan is executed, and (3) directing a modification the well site operation plan using the analyzed gravitational parameters and the subterranean formation parameters.

Description

BRIEF DESCRIPTION

(1) Reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is an illustration of a diagram of an example well site with a production well and an injection well;

(3) FIG. 2 is an illustration of a diagram of an example well site demonstrating an injection fluid flow detection;

(4) FIG. 3 is an illustration of a diagram of an example graph demonstrating a water front detection;

(5) FIG. 4 is an illustration of a diagram of an example well site demonstrating orphaned hydrocarbon reservoir detection;

(6) FIG. 5 is an illustration of a flow diagram of an example method to collect and analyze gravitational data;

(7) FIG. 6 is an illustration of a block diagram of an example quantum gravimeter data analyzer system; and

(8) FIG. 7 is an illustration of a block diagram of an example quantum gravimeter data analyzer controller according to the principles of the disclosure.

DETAILED DESCRIPTION

(9) In the hydrocarbon production industry, reservoirs can be identified having potential hydrocarbons in subterranean formations below the surface. Developing the reservoir can be performed using various techniques, such as drilling one or more wellbores, injecting fluid, such as water, brine, or carbon dioxide into a wellbore, hydraulic fracturing (HF), or using other techniques. As fluids, gasses, brine, or water (collectively, fluids) is pumped downhole, or as hydrocarbons or other subterranean formation material is extracted from a subterranean formation region (e.g., moving fluids into or out of the wellbore), the mass distribution of material and fluids in the subterranean formation can change. This can be due to the injection or extraction of material and fluids, or from a change in distribution of material or fluids within the subterranean formation.

(10) The change in mass distribution in the subterranean formation can be measured using gravimeters, collecting gravitational data due to changes in the gravitational gradient. Conventional gravimeters used in hydrocarbon production industries can be located at a surface location and can be subject to calibration errors over time, such as drift. Being able to improve the accuracy of the collected gravitational data to produce a more accurate representation of the various fluid flows in a subterranean formation would be beneficial. It can improve the drainage of a subterranean formation reservoir while minimizing the amount of waste water extracted with the hydrocarbons.

(11) This disclosure presents processes to improve the measurement of gravitational gradient changes over time as well sites are developed or hydrocarbons are produced. Knowing how close a water front, such as from an underground water source, is to a wellbore can be used to determine if specified zones of the wellbore should be closed. Existing sensors can detect water after it enters the wellbore, while this disclosure demonstrates a way to detect the water at a distance away from the wellbore where corrective action can be performed. For example, diverter material can be delivered to specified fractures in a HF wellbore to close them off to prevent water from entering the wellbore, or valves can be opened or closed at an injection wellbore, or zones can be closed in a production wellbore.

(12) In some aspects, the disclosed processes can measure carbon sequestration. Carbon sequestration can be observed by measuring the carbon dioxide going into the subterranean formation and the carbon dioxide that remains in the subterranean formation. In some aspects, carbon sequestration can include measuring the amount of carbon dioxide that chemically reacts downhole, such as through a process called mineral carbonation to form rock-like materials.

(13) The gravitational gradient changes can be detected, in part, since the density of gases, water, and oil are different from each other. Changes in the mass of one of the fluids can change the gravitational gradients. For example, in a demonstration well site, when 1000 barrels per day (BPD) is produced, then the daily mass change can be approximately a half million pounds. This mass change can cause the changes in the gravitational gradient. The collected gravitational data can represent an absolute gravity reading, a relative gravity reading, or a combination of gravity readings. The gravity measurements can include magnitude and direction.

(14) Quantum gravimeters can be used to collect the gravitational data. A quantum gravimeter is a matter-wave interferometer that is sensitive to small changes in gravitational attraction. In some aspects, quantum gravimeters can use rubidium atoms that are cooled by lasers to just above absolute zero. In some aspects, the quantum gravimeters can cool the measuring atoms, such as rubidium, to less than 10 Kelvin. The cloud of atoms is propelled upward in a high vacuum and then measured as they fall back under gravity.

(15) One of the advantages of a quantum gravimeter is that it has negligible drift. The drift can be less than 10 micro Galileos per day and can be less than 0.1 micro Galileos per day. A Galileo is a unit of measuring gravity and equals one milli-g. For example, the Earth's average gravity is 981 Galileos. Another unit is the Eotvos which is the gravity gradient. An Eotvos is a nano Galileo per

centimeter. An average sized adult human who is two meters away can produce approximately one Eotvos of gravity gradient towards them.

(16) Another advantage of a quantum gravimeter is increased sensitivity as compared to non-quantum gravimeters. Quantum gravimeters can have accuracy greater than 10 micro Galileos and can have accuracy finer than 1 micro Galileo. As a result, gradients finer than 10 Eotvos can be obtained and many finer than 0.1 Eotvos or even 1 milli Eotvos can be determined.

(17) There are two types of gravimeters: absolute gravity sensors and gravity gradient sensors. A relative gravimeter is a type of gravity gradient sensor. The gravimeters can be scalar measuring an amplitude, one directional, or three-directional. For surface applications, there can be two types of installations: fixed gravimeters and moving gravimeters, e.g., mobile gravimeters.

(18) At least one quantum gravimeter is present at a reservoir. In some aspects, the quantum gravimeter can be located proximate an injection well site (e.g., an injection well system), a HF well site (e.g., a HF well system), a production well site (e.g., a production well system), or an observation well site (e.g., an observation well system). The reservoir or wellsite within the reservoir area can have more than one well site or well site type. For example, there could be an injector/producer combination, with one or more injector wellbores and one or more producer wellbores. In some aspects, the injector and producer wellbores can be the same wellbore. In some aspects, HF wellbores can be used in various combinations with one or more injector and producer wellbores.

(19) In some aspects, the fixed quantum gravimeter can be fixedly attached at the surface. In some examples, it can be located near the wellbore such as in the yard. In other examples, it can be a distance from the wellbore, remaining proximate the wellsite, so that it can register ground water changes and can be less sensitive to hydrocarbon flow. The fixed quantum gravimeter can provide a continuous measurement of the gravitational acceleration (or, alternatively, gravitational gradient). The continuous measurements can include a measurement once every minute, day, week, or other collection time interval. The measurements can allow the fixed quantum gravimeter to distinguish between the slow formation drain versus the faster changes from tides, surface equipment, or water table changes.

(20) The quantum gravimeter can have a significantly lower amount of drift as compared to conventional, non-quantum gravimeters, currently employed. This can improve the accuracy of the gravitational data collected. In some aspects, the gravitational data collected can be utilized as input data to calibrate non-quantum gravimeters located at the same reservoir area. In some aspects, gravitational data from a fixed quantum gravimeter can be utilized by a user or be automatically used to calibrate a mobile non-quantum gravimeter.

(21) In some aspects, the quantum gravimeter can be located at a surface location proximate to the reservoir. In some aspects, there can be more than one gravimeter at a surface location, such as a mix of quantum gravimeters and non-quantum gravimeters. In some aspects, the non-quantum gravimeters can be MEMs-based, accelerometer, or resonant spring gravimeters. In some aspects, all of the surface gravimeters can be quantum gravimeters. In some aspects, one or more of the surface gravimeters can be mobile, meaning that a gravimeter can collect gravitational data for a specified integration time interval then move to another location on the surface and collect additional gravitational data at the subsequent collection time interval.

(22) In some aspects, the gravimeters, whether a quantum gravimeter or a non-quantum gravimeter, can collect gravitational data over an integration time interval. One set of gravitational data can be collected over the integration time interval, such as 1 second, 100 seconds, 1,000 seconds, or other smaller or larger time intervals. Increased resolution can be achieved by increasing the integration time interval for the measurement, e.g., the longer the integration time interval can improve the accuracy of the collected gravitational data to a certain limit. Too long of an integration time interval could cause a blurring of the gravitational data as fluid flows into or out of the subterranean formation region being measured.

(23) In some aspects, the gravimeters proximate to the reservoir can continuously collect gravitational data, for each integration time interval. Changes to the gravitational gradient may not change rapidly, so continuously collecting gravitational data can impose constraints on efficiency, power consumption, or data storage. Therefore, in some aspects, a collection time interval can be utilized where the gravimeters collect gravitational data at specified times or after an elapse of the collection time interval. The collection time interval can be one minute, one hour, one day, or shorter or longer time periods as specified by the well site operation plan or reservoir operation plan.

(24) In some aspects, a gravity gradient can be measured using two gravimeters that are at a fixed distance apart. In some aspects, three gravimeters can be used, at fixed separation distances from each other, to determine changes in the gravity gradient. This type of gravimeter system can estimate the distance to the subterranean mass that is changing over time.

(25) One challenge from a surface gravitational mapping is that it can be uncertain at what depth the gravity is changing. The surface gravitational map can be subject to noise from the surface. For example, changes in a water table can produce gravitational changes that can appear very large due to the closeness of the water table versus the farther distance to hydrocarbon formations. Similarly, soil subsidence, tidal effects, barometric pressure, or vehicle movement can cause larger signals than the downhole fluid movement.

(26) The downhole fluid movement can be determined using the gravity measurement, the gravity gradient measurement, or a combination thereof. The gravity gradient can provide an estimate for the distance from the mass. The magnitude of gravity decreases by $1/R^2$, where R is the distance to the mass. Thus, the rate of change of the gravity, e.g., the gravity gradient, can be used to estimate the distance to the mass. If the mass change is a water table, such as in the above example, then the $1/R^2$ change is much greater because the distance is much less than a more distant subterranean mass, such as an oil reservoir.

(27) To compensate for the uncertainty of depth, one or more downhole gravimeters, i.e., a set of downhole gravimeters, can be used in combination with the one or more surface gravimeters, i.e., a set of surface gravimeters. The combination can provide an improved representation of the fluid and mass changes occurring in the subterranean formation, such as fluid movement, and can be used to reduce gravimetric noise. For example, gravimetric noise can consist of changes in the water table due to flood or drought conditions. The fixed surface gravimeter can determine the gravimetric effects of the water table changes by noting the rate of change of the gravimetric changes and correlating the gravimetric changes with rain fall as well as direct water table measurements, where the water table measurements or weather/environmental parameters can be used as input parameters to the disclosed methods. Analysis of the gravitational data from the fixed quantum gravimeter can have trouble determining the extent of the reservoir remaining and whether any sections of trapped hydrocarbons were left behind. A survey, including the fixed quantum gravimeter and a set of downhole or mobile gravimeters can allow mapping of these spatially distributed areas.

(28) In some aspects, a downhole gravimeter can be used in conjunction with the surface quantum gravimeter, and other surface gravimeters if being used. The gravitational data collected by the downhole gravimeter can be used in the processing of gravitational data to determine the analyzed gravitational parameters and subterranean formation parameters. In some aspects, there can be more than one downhole gravimeter. The downhole gravimeters can utilize one or more different collection time intervals or integration time intervals, and need not be the same as the collection time interval or integration time interval utilized by the surface quantum gravimeter.

(29) If the downhole gravimeter is a non-quantum gravimeter, then one or more of the surface quantum gravimeters can be used to calibrate the downhole gravimeter. In some aspects, one or more of the set of downhole gravimeters can be a quantum gravimeter. The gravimeters in the set of downhole gravimeters can be placed at fixed locations along the wellbore, for example, no

closer than a length of pipe segment, 20,000 feet apart, or other distances can be used. In some aspects, the gravimeters in the set of downhole gravimeters have a lesser sensitivity than the gravimeters used at the surface locations. This is due to the environmental conditions experienced downhole that could adversely impact the type of gravimeter used downhole and the effectiveness of its operations. Conventional gravimeters can drift by 0.1 milliGalileos per day at room temperature. At downhole temperatures, the drift of a conventional gravimeter can exceed 1 milliGalileos per day.

(30) In some aspects, a combination of multiple gravity measurements can allow for a better mapping of the movement of downhole fluids by using a gravitational numerical model to minimize gravitational noise. In some aspects, an inversion process can be applied to the gravitational data to remove noise.

(31) In some aspects, the gravitational data can be communicated to a computing system, a well site controller, a reservoir controller, a data center, a cloud environment, or other well site equipment or distant computing system. The gravitational data from one or more quantum gravimeters and zero or more non-quantum gravimeters can be processed and analyzed.

(32) In some aspects, the result of the processing can be analyzed gravitational parameters, for example, gravitational gradients, calibration parameters for non-quantum gravimeters, and other types of gravitational parameters. In some aspects, the result of the processing can be subterranean formation parameters, for example, localizing a water table, localizing a water front, identifying orphaned hydrocarbon fields, determining a size and location of a remaining hydrocarbon reservoir, determining the flow of injected fluid or HF slurries, identifying damage, or other features or characteristics of the subterranean formation.

(33) In some aspects, the subterranean formation parameters can specify one or more mass changes in one or more parts or regions of the subterranean formation. In some aspects, the analyzed gravitational parameters can specify a determined depth parameter of a fluid in the subterranean formation. In some aspects, the analyzed gravitational parameters can specify a determined direction parameter to a fluid in the subterranean formation. In some aspects, the analyzed gravitational parameters can specify a determined fluid movement parameter of a fluid flow or a gravimetric noise parameter. In some aspects, the gravimetric noise parameter can be determined using a gravitational numerical model.

(34) In some aspects, the subterranean formation parameters can specify a determined subterranean formation damage parameter, for example, a deposition of solids during injection or a scale deposition during production. A damage parameter can be used to describe the amount of deposition at a location of the wellbore and thereby provide a warning of a potential stuck pipe scenario. A well site operation plan or a reservoir operation plan can be updated or modified using the analyzed gravitational parameters or the subterranean formation parameters.

(35) Turning now to the figures, FIG. 1 is an illustration of a diagram of an example well site **100** with a production well and an injection well. In other aspects, there can be one or more additional well sites, such as production wells, injection wells, observation wells, and other types of well sites. A surface gravimeter can be utilized to map how the gravitational attraction changes over time.

(36) The solid lines in the contour plots represent the gravity as measured from the surface. The gravitational acceleration can vary, for example, by 2 milliGalileos to 100s of milliGalileos. Over time, the gravitational acceleration can change as fluids are produced and fluids are injected. The contour lines of the gravitational acceleration can vary after a time period of well site or reservoir operations. Gravitational attraction can be decreased proximate the production well and can be increased proximate the injection well.

(37) Well site **100** includes a production well site **110** and an injection well site **115**. Solid lines **120** proximate production well site **110** indicate the gravitational gradient prior to fluid being injected in the reservoir by injection well site **115**. Dashed lines **122** proximate production well site **110**

indicate the gravitational gradient after fluid has been injected by injection well site **115**. Likewise, solid line **130** proximate injection well site **115** indicates the gravitational gradient prior to fluid injection, and dashed line **132** proximate injection well site **115** indicates the gravitational gradient after fluid injection. The quantum gravimeter and zero or more additional gravimeters of various types can be used, after data analysis, to detect the gravitational gradient movement from solid lines **120** to dashed lines **122**, and solid line **130** to dashed line **132**.

(38) FIG. 2 is an illustration of a diagram of an example well site **200** demonstrating an injection fluid flow detection. The challenge from a surface gravitational mapping is that it can be uncertain at what depth the fluids are moving. If the injected fluids are short-circuiting to the production well, then the gravitational contour map can have difficult identifying that the fluids are moving at a single zone rather than moving across the entire formation. Additionally, the surface gravitational map can be subject to noise from the surface. For example, changes in the water table can produce gravitational changes that can appear very large due to the closeness of the water table and the farther distance to hydrocarbon formation.

(39) Depth information can be obtained by combining the surface gravimetry with downhole gravimetry. The downhole gravimeter can be fixedly attached to the downhole tubing, preferably as part of the downhole completion that would also include screens and possibly a flow restrictor. The gravimeter can be part of the injector, the producer, or both, or as a separate observation well. Multiple downhole gravimeters can be part of the completion and can communicate their respectively collected measurements to the surface through conventional communication techniques.

(40) The signal from the non-moving downhole gravimeters can be combined with the surface gravimeter to map the depth of the fluid movement as well as to improve the distinguishment between fluid movement in the reservoir and gravimetric noise. For example, a well site which has high-density water being injected and lower density hydrocarbons being produced, and where a water table exists near the surface, can cause gravimetric noise. Gravimetric noise can consist of changes in the water table due to flood or drought conditions.

(41) Well site **200** includes a derrick **205** at a surface **206**, a well site controller **207**, and a computing system **208**. Well site controller **207** can be positioned central to the well site operation or local to the one or more equipment devices to form a data network among other equipment devices or data transmitters. Well site controller **207** includes a processor and a memory, and is configured to direct operation of well site **200**. Derrick **205** is located at a surface **206**.

(42) Extending below derrick **205** is a wellbore **210** with a fluid pipe **215** positioned within wellbore **210**. Fluid pipe **215** is shown as ending with a dashed pipe indicating that fluid pipe **215** can extend to various lengths, with a maximum length being the length of wellbore **210**. There can be downhole tools located at one or more locations within wellbore **210**, for example at an end of fluid pipe **215**, at one or more locations along a casing, or at one or more locations along wellbore **210** where there is no casing. The downhole tools can include various tools, such as sensors, pumps, and other tools. Other components of downhole tools can be present, such as a local power supply (e.g., generators, batteries, or capacitors), telemetry systems, transceivers, and control systems. Wellbore **210** is surrounded by subterranean formation **250**.

(43) Well site controller **207** or computing system **208** which can be communicatively coupled to well site controller **207**, can be utilized to communicate with the downhole tools, such as sending and receiving telemetry, data, instructions, subterranean formation measurements, gravitational data, and other information. Computing system **208** can be proximate well site controller **207** or be a distance away, such as in a cloud environment, a data center, a lab, or a corporate office. Computing system **208** can be a laptop, smartphone, PDA, server, desktop computer, cloud computing system, other computing systems, or a combination thereof, that are operable to perform the processes described herein. Well site operators, engineers, and other personnel can send and receive data, instructions, measurements, and other information by various conventional means,

now known or later developed, with computing system **208** or well site controller **207**. Well site controller **207** or computing system **208** can communicate with the downhole tools using conventional means, now known or later developed, to direct operations of the downhole tools.

(44) Well site **200** includes an injection well **260** with a pumping system **265**. Injection well **260** is pumping an injection fluid **285** downhole wellbore **270**. The arrows proximate wellbore **270** indicate that injection fluid **285** enters subterranean formation **250** at various locations.

(45) At surface **206**, there is a quantum gravimeter **220**. In some aspects, more than one quantum gravimeter can be present at surface **206**, or other types of gravimeters can be present. A water table **280** is present in this example. Quantum gravimeter **220** can be affected by gravitational noise from water table **280** due to the proximity of water table **280** to quantum gravimeter **220**. The gravitational noise could affect the accuracy of the analyzed gravitational parameters and subterranean formation parameters leading to a misrepresentation of an injection fluid **285**, which is shown as extending close to wellbore **210**.

(46) To improve detection of this extension of injection fluid **285**, and thereby allowing corrective actions to be conducted, such as closing specific valves or zones in wellbore **270** or in wellbore **210**, additional gravimeters can be located downhole. At fixed locations within wellbore **210** are a gravimeter **222a**, a gravimeter **222b**, a gravimeter **222c**, and a gravimeter **222d**, collectively, a set of downhole gravimeters **222**. Set of downhole gravimeters **222** can be one or more of various types of gravimeters, such as quantum gravimeters, MEMs gravimeters, or other types of gravimeters. The gravitational data collected by set of downhole gravimeters **222**, combined with the gravitational data collected by quantum gravimeter **220** can produce an improved accuracy of the analyzed gravitational parameters and subterranean formation parameters. In this example, an improved detection of the finger extension of injection fluid **285** can be produced, such as improved accuracy of depth and direction parameters.

(47) FIG. **3** is an illustration of a diagram of an example graph **300** demonstrating a water front detection. Graph **300** is demonstrating a time trace of the collected gravimetric data, where relative changes in the gravimetric data can indicate what is happening in the subterranean formation. Changes in the water table, such as a higher water table after heavy rain, can have a greater change at the surface than deep downhole a wellbore and can be detected where the gravimetric data changes in opposite directions. Each of the gravimeter sensors can have a different baseline gravitational acceleration that can be correlated to the depth from the surface.

(48) Graph **300** has an x-axis **305** which shows the progression of time. A y-axis **306** indicates the depth from a surface location. Y-axis **306** identifies a surface quantum gravimeter and the relative depths of four downhole gravimeters. For example, the surface quantum gravimeter can be quantum gravimeter **220**, and the four downhole gravimeters can be set of downhole gravimeters **222**. The surface quantum gravimeter collects gravitational data used to plot line **320a**. The first downhole gravimeter collects gravitational data used to plot line **320b**. The second downhole gravimeter collects gravitational data used to plot line **320c**. The third downhole gravimeter collects gravitational data used to plot line **320d**. The fourth downhole gravimeter collects gravitational data used to plot line **320e**.

(49) The finger extension of injection fluid **285** can be first detected by the surface quantum gravimeter and the fourth gravimeter, gravimeter **222d**, since these gravimeters are closer to injection well **260**. The relative strength of the analyzed gravitational parameters between the surface quantum gravimeter and the fourth gravimeter can provide an estimate of the depth of the fluid change.

(50) The relative changes between the gravitational data collected from the downhole gravimeters can provide indicators for the depth of the fluid movement. Stronger changes can be detected by the gravimeters closer to the fluid flow and the relative magnitude of the change can indicate a relative position.

(51) Dashed line **330** indicates a time when a high water table, such as water table **280**, has been

detected. A dashed line **332** indicates a time when the high water table has been compensated for in the analysis of the gravitational data. A dashed line **334** indicates a time when a water front, such as the finger extension of injected fluid **285**, is first detected. At dashed line **334**, analysis of plot line **320a** shows a beginning of a trend upwards and analysis of plot line **320e** shows a beginning of a trend downwards.

(52) At a dashed line **336**, the water front is continuing to approach the wellbore, approaching the downhole gravimeters. At dashed line **336**, analysis of plot line **320a** and plot line **320e** shows a steeper change in the gravitational gradient, relative to the analysis at dashed line **334**. Analysis of plot line **320b**, plot line **320c**, and plot line **320d** demonstrate a smaller trend in a respective of an upwards or downwards direction. The upwards or downwards direction can assist in deriving the depth of the water front.

(53) The surface gravimetric data can be important for determining the direction from which the fluids are moving. The fixed downhole readings can provide an indication of depth while being less accurate on the direction. The surface gravimetric data can be used to determine an indication of direction of the water front while being less accurate to estimate its depth. The combination of surface and downhole gravimeters can be utilized to improve the analysis of the gravitational data and improve the accuracy of water front detections, hydrocarbon detections, and other fluid detections.

(54) The downhole gravimeters can have lesser sensitivity than the surface gravimeters as they are closer to the moving fluids and having a high-sensitivity, high-temperature, small-size, non-replaceable, low-power downhole sensor can be difficult.

(55) Graph **300** is demonstrated using absolute gravity readings. A similar result can be obtained utilizing relative gravity readings, or utilizing a combination of absolute and relative gravity readings. Graph **300** uses the injector/producer model as shown in well site **200** of FIG. 2. In some aspects, there can be multiple injectors, multiple producers, no injectors, a single well that is used for injection as well as for production, or various combinations thereof. HF well sites are a combination of injection and production well systems and the disclosed processes can be used in that type of well system.

(56) FIG. 4 is an illustration of a diagram of an example well site **400** demonstrating orphaned hydrocarbon reservoir detection. Well site **400** utilizes the same production well system as shown in well site **200** of FIG. 2. Set of downhole gravimeters **222** are positioned similarly as in well site **200**. Overlaid set of downhole gravimeters **222** and wellbore **210** is an original hydrocarbon reservoir **420**.

(57) During the production operation stages, original hydrocarbon reservoir **420** has been reduced through the removal of hydrocarbons up through wellbore **210**. At a future production operation stage, original hydrocarbon reservoir **420** has shrunk as shown by hydrocarbon reservoir **425**. In this demonstration, there are two orphaned hydrocarbon reservoirs, shown as hydrocarbon reservoir **430** and hydrocarbon reservoir **432**. These types of fluid movement, reservoir sizing, reservoir location, orphan reservoir identification, and other reservoir fluid changes can be detected by analyzing the collected gravitational data from the surface quantum gravimeter and the one or more downhole gravimeters.

(58) FIG. 5 is an illustration of a flow diagram of an example a method **500** to collect and analyze gravitational data. Method **500** can be performed, for example, by users performing analysis operations. Some steps of method **500** can be performed on a computing system, for example, quantum gravimeter data analyzer system **600** of FIG. 6 or quantum gravimeter data analyzer controller **700** of FIG. 7. The computing system can be a reservoir controller, a well site controller, a data center, a cloud environment, a server, a laptop, a mobile device, or other computing system capable of receiving the gravitational data, input parameters, and capable of communicating with other computing systems. Some steps of method **500** can be encapsulated in software code or in hardware, for example, an application, code library, dynamic link library, module, function, RAM,

ROM, and other software and hardware implementations. The software can be stored in a file, database, or other computing system storage mechanism. Some steps of method **500** can be partially implemented in software and partially in hardware.

(59) Method **500** starts at a step **505** and proceeds to a step **510**. In step **510**, a quantum gravimeter can be located at a surface location of a well site proximate a hydrocarbon reservoir. In some aspects, one or more additional gravimeters can be positioned at the surface, where the gravimeters can be quantum gravimeters, non-quantum gravimeters, or a combination thereof. In a step **515**, zero or more gravimeters, which can be various combinations of quantum gravimeters and non-quantum gravimeters, can be located at positions downhole a wellbore of the well site.

(60) In a step **520**, gravitational data can be collected from each of the active gravimeters located at the surface proximate the well site or downhole a wellbore of the well site. The data can be collected over an integration time interval, where a longer integration time interval can generally improve the resolution of the gravitational data collected. An excessively long integration time interval could begin to cloud or blur the data as subterranean formation fluids shift in real time. The integration time interval can be the same, different, or a combination thereof across each of the active gravimeters.

(61) Each gravimeter can collect gravitational data over its respectively specified integration time interval at each collection time interval. Depending on the operation plan stage, the collection time interval can be modified. For example, during active injection stages, the collection time interval can be smaller, such as one minute, one hour, or another time interval. During a non-active injection stage, the collection time interval can be larger, such as a day, three days, or a week, where the natural flow of subterranean formation fluids can be measured.

(62) In a step **525**, a computing system can receive the collected gravitational data from one or more of the active gravimeters. The computing system can be proximate the well site or distant from it. For example, the computing system can be a reservoir controller, a well site controller, a server, a laptop, a data center, a cloud environment, an edge computing system, or other type of computing systems. The gravitational data can be analyzed, using each set of received gravitational data. In some aspects, a pre-processing can be applied to the gravitational data, for example, an inversion process. In some aspects, noise reduction algorithms can be applied. In some aspects, a machine learning system or a deep neural network system can be utilized to process and analyze the gravitational data. A machine learning or deep neural network system can apply learnings from other gravitational data collections at this well site or other well sites to further improve noise reduction and improve the resulting analysis.

(63) In a step **530**, the computing system can generate the analyzed gravitational parameters and the subterranean formation parameters, i.e., the results. The analyzed gravitational parameters can include, for example, gravitational gradients, calibration parameters for non-quantum gravimeters, and other types of gravitational parameters. The subterranean formation parameters can include, for example, localizing a water table, localizing a water front, identifying orphaned hydrocarbon fields, determining a size and location of a remaining hydrocarbon reservoir, determining the flow of injected fluid or HF slurries, identifying damage, or other features or characteristics of the subterranean formation.

(64) In a step **535**, the analyzed gravitational parameters and subterranean formation parameters can be communicated to one or more systems. For example, the results can be communicated to a well site operation plan system within a well site controller. The results can be communicated to another system or equipment proximate the well site or distant from the well site.

(65) Method **500** can optionally proceed to a step **540**, to a step **545**, or a decision step **550**. Step **540**, step **545**, or decision step **550** can be performed in serial, parallel, overlapping, or various combinations thereof, and in various orders. In aspects, where step **540** and step **545** are not performed, then method **500** proceeds to decision step **550**.

(66) In step **540**, the analyzed gravitational parameters and the subterranean formation parameters

can be utilized to modify a well site operation plan or a reservoir operation plan. Method **500** proceeds to decision step **550**. In step **545**, if mobile gravimeters are being utilized, then they can be moved to their next designated location. Method **500** proceeds to decision step **550**.

(67) In decision step **550**, a determination can be made whether the method ends or method **500** continues to step **520** after the next collection time interval. The method can end due to various factors, such as a change in the stage of a well operation plan or a reservoir operation plan, or another end of method event as specified by the respective operation plan or a user. If decision step **550** resultant is 'Yes', method **500** proceeds to step **520**. If decision step **550** resultant is 'No', method **500** proceeds to a step **595** and the method ends.

(68) FIG. **6** is an illustration of a block diagram of an example quantum gravimeter data analyzer system **600**, which can be implemented in one or more computing systems, for example, a data center, cloud environment, server, laptop, smartphone, tablet, and other computing systems. In some aspects, quantum gravimeter data analyzer system **600** can be implemented using a gravimeter data analyzer controller such as quantum gravimeter data analyzer controller **700** of FIG. **7**. Quantum gravimeter data analyzer system **600** can implement one or more steps of one or more methods of this disclosure, such as method **500** of FIG. **5**.

(69) Quantum gravimeter data analyzer system **600**, or a portion thereof, can be implemented as an application, a code library, a dynamic link library, a function, a module, other software implementation, or combinations thereof. In some aspects, quantum gravimeter data analyzer system **600** can be implemented in hardware, such as a ROM, a graphics processing unit, or other hardware implementation. In some aspects, quantum gravimeter data analyzer system **600** can be implemented partially as a software application and partially as a hardware implementation. Quantum gravimeter data analyzer system **600** is a functional view of the disclosed processes and an implementation can combine or separate the described functions in one or more software or hardware systems.

(70) Quantum gravimeter data analyzer system **600** includes a data transceiver **610**, a gravimeter data analyzer **620**, and a result transceiver **630**. The results, e.g., the analyzed gravitational parameters, the subterranean formation parameters, and interim outputs from gravimeter data analyzer **620** can be communicated to a data receiver, such as one or more of a user or user system **660**, a well site controller **662**, a reservoir controller **664**, a computing system **666**, or other processing or storage systems **668**. The results can be used to determine changes or modifications that may be needed to a well site operation plan or a reservoir operation plan, for example, adjusting where and when to inject an injection fluid, the composition of the injection fluid, or opening or closing valves or zones within the wellbore.

(71) Data transceiver **610** can receive input parameters, such as parameters to direct the operation of the analysis implemented by gravimeter data analyzer **620**. For example, an input parameter can be the integration time interval, collection time interval, or location for each gravimeter being utilized. In some aspects, data transceiver **610** can be part of gravimeter data analyzer **620**.

(72) Result transceiver **630** can communicate one or more results or interim outputs to one or more data receivers, such as user or user system **660**, well site controller **662**, reservoir controller **664**, computing system **666**, other processing or storage systems **668**, or other related systems, whether located proximate result transceiver **630** or distant from result transceiver **630**. Data transceiver **610**, gravimeter data analyzer **620**, and result transceiver **630** can be, or can include, conventional interfaces configured for transmitting and receiving data. In some aspects, gravimeter data analyzer **620** can be a machine learning system, for example, a deep neural network system or a machine learning algorithm. In some aspects, one or more of data transceiver **610**, gravimeter data analyzer **620**, or result transceiver **630** can be part of the machine learning system.

(73) Gravimeter data analyzer **620** can implement the analysis and algorithms as described herein utilizing the gravitational data and the input parameters. For example, gravimeter data analyzer **620** can perform the noise reduction of the received gravitational data, analyze the gravitational data,

and generate the analyzed gravitational parameters and the subterranean formation parameters. A memory or data storage of gravimeter data analyzer **620** can be configured to store the processes and algorithms for directing the operation of gravimeter data analyzer **620**. Gravimeter data analyzer **620** can also include a processor, such as processor **730** of FIG. 7, that is configured to operate according to the analysis operations and algorithms disclosed herein, and an interface to communicate (transmit and receive) data.

(74) FIG. 7 is an illustration of a block diagram of an example quantum gravimeter data analyzer controller **700** according to the principles of the disclosure. Quantum gravimeter data analyzer controller **700** can be stored on a single computer or on multiple computers. The various components of quantum gravimeter data analyzer controller **700** can communicate via wireless or wired conventional connections. A portion or a whole of quantum gravimeter data analyzer controller **700** can be located at one or more locations and other portions of quantum gravimeter data analyzer controller **700** can be located on a computing device or devices located downhole or at a surface location. In some aspects, quantum gravimeter data analyzer controller **700** can be wholly located at a surface or distant location. In some aspects, quantum gravimeter data analyzer controller **700** can be part of another system, and can be integrated in a single device, such as a part of downhole tools, or part of a well site controller or reservoir controller.

(75) Quantum gravimeter data analyzer controller **700** can be configured to perform the various functions disclosed herein including receiving input parameters and gravitational data, and generating results from an execution of the methods and processes described herein, such as analyzed gravitational parameters and subterranean formation parameters. Quantum gravimeter data analyzer controller **700** includes a communications interface **710**, a memory **720**, and a processor **730**.

(76) Communications interface **710** is configured to transmit and receive data. For example, communications interface **710** can receive the input parameters and the gravitational data. Communications interface **710** can transmit the generated results, data from the input files, the gravitational data, or interim outputs. In some aspects, communications interface **710** can transmit a status, such as a success or failure indicator of quantum gravimeter data analyzer controller **700** regarding receiving the various inputs, transmitting the generated results, or producing the generated results.

(77) In some aspects, communications interface **710** can receive input parameters from a machine learning system, for example, where the gravitational data can be processed for noise reduction or data clarity improvement, such as is described for gravimeter data analyzer **620**.

(78) In some aspects, the machine learning system can be implemented by processor **730** and perform the operations as described by gravimeter data analyzer **620**. In some aspects, processor **730** can implement a process such as a computer program product having a series of operating instructions stored on a non-transitory computer-readable medium that directs a data processing apparatus when executed thereby to perform operations to generate analyzed gravitational parameters and subterranean formation parameters, the operations can include (1) collecting gravitational data at the one or more surface quantum gravimeters, (2) processing the gravitational data received from the one or more surface quantum gravimeters, generating analyzed gravitational parameters and subterranean formation parameters that specify mass changes of a subterranean formation of the reservoir as a well site operation plan stage is executed, and (3) modifying the well site operation plan using the analyzed gravitational parameters and the subterranean formation parameters.

(79) Communications interface **710** can communicate via communication systems used in the industry. For example, wireless or wired protocols can be used. Communication interface **710** is capable of performing the operations as described for data transceiver **610** and result transceiver **630** of FIG. 6.

(80) Memory **720** can be configured to store a series of operating instructions that direct the

operation of processor **730** when initiated, including the code representing the algorithms for determining processing the collected data. Memory **720** is a non-transitory computer readable medium. Multiple types of memory can be used for data storage and memory **720** can be distributed.

(81) Processor **730**, which can be a gravitational processor, can be configured to produce the results, one or more interim outputs, or statuses utilizing the received inputs. Processor **730** can be configured to direct the operation of the quantum gravimeter data analyzer controller **700**.

Processor **730** includes the logic to communicate with communications interface **710** and memory **720**, and perform the functions described herein. Processor **730** is capable of performing or directing the operations as described by gravimeter data analyzer **620** of FIG. **6**.

(82) A portion of the above-described apparatus, systems or methods may be embodied in or performed by various analog or digital data processors, wherein the processors are programmed or store executable programs of sequences of software instructions to perform one or more of the steps of the methods. A processor may be, for example, a programmable logic device such as a programmable array logic (PAL), a generic array logic (GAL), a field programmable gate arrays (FPGA), or another type of computer processing device (CPD). The software instructions of such programs may represent algorithms and be encoded in machine-executable form on non-transitory digital data storage media, e.g., magnetic or optical disks, random-access memory (RAM), magnetic hard disks, flash memories, and/or read-only memory (ROM), to enable various types of digital data processors or computers to perform one, multiple or all of the steps of one or more of the above-described methods, or functions, systems or apparatuses described herein.

(83) Portions of disclosed examples or embodiments may relate to computer storage products with a non-transitory computer-readable medium that have program code thereon for performing various computer-implemented operations that embody a part of an apparatus, device or carry out the steps of a method set forth herein. Non-transitory used herein refers to all computer-readable media except for transitory, propagating signals. Examples of non-transitory computer-readable media include, but are not limited to: magnetic media such as hard disks, floppy disks, and magnetic tape; optical media such as CD-ROM disks; magneto-optical media such as floppy disks; and hardware devices that are specially configured to store and execute program code, such as ROM and RAM devices. Examples of program code include both machine code, such as produced by a compiler, and files containing higher level code that may be executed by the computer using an interpreter.

(84) In interpreting the disclosure, all terms should be interpreted in the broadest possible manner consistent with the context. In particular, the terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced.

(85) Those skilled in the art to which this application relates will appreciate that other and further additions, deletions, substitutions and modifications may be made to the described embodiments. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the present disclosure will be limited only by the claims. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. Although any methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the present disclosure, a limited number of the exemplary methods and materials are described herein.

(86) Each of the disclosed aspects in the SUMMARY can have one or more of the following additional elements in combination. Element 1: further comprising modifying the well site operation plan using the analyzed gravitational parameters and the subterranean formation parameters. Element 2: further comprising communicating the analyzed gravitational parameters and the subterranean formation parameters to a well site controller, a reservoir controller, a

computing system, a data center, or a cloud environment. Element 3: wherein the well site is an injection well system and a first fluid is pumped downhole one or more of the one or more wellbores. Element 4: wherein the well site is a production well system and a second fluid is extracted from one or more of the one or more wellbores. Element 5: wherein the well site is a HF well system, and a HF fluid is pumped downhole one or more of the one or more wellbores. Element 6: wherein the well site is an observation well system. Element 7: wherein the collecting occurs at a collection time interval. Element 8: wherein the gravitational data represents one of an absolute gravity reading, a relative gravity reading, or a combination of gravity readings. Element 9: wherein one or more non-quantum gravimeters are proximate the well site. Element 10: further comprising calibrating the one or more non-quantum gravimeters using the gravitational data received from the one or more surface quantum gravimeters. Element 11: wherein one or more downhole gravimeters are located downhole one or more of the one or more wellbores. Element 12: wherein the processing uses gravitational data from the one or more downhole gravimeters. Element 13: wherein the processing determines a depth parameter of a subterranean fluid in the subterranean formation. Element 14: wherein the processing determines a direction parameter to a subterranean fluid in the subterranean formation. Element 15: wherein the processing determines a fluid movement parameter or a gravimetric noise parameter. Element 16: wherein the gravimetric noise parameter is determined using a gravitational numerical model. Element 17: wherein the processing determines a subterranean formation damage parameter. Element 18: wherein the one or more of the one or more downhole gravimeters are downhole quantum gravimeters. Element 19: wherein the one or more of the one or more downhole gravimeters are non-quantum gravimeters. Element 20: wherein the one or more downhole gravimeters are at respective fixed locations within the one or more wellbores. Element 21: wherein the one or more downhole gravimeters have lesser sensitivity than the one or more surface quantum gravimeters. Element 22: wherein the collecting further comprises receiving input parameters at one or more of the one or more surface quantum gravimeters. Element 23: wherein the input parameters specify one or more of an integration time interval or a collection time interval. Element 24: wherein the integration time interval is in a range of 1 second to 1,000 seconds. Element 25: wherein a first set of the one or more surface quantum gravimeters are at respective fixed locations and a second set of the one or more surface quantum gravimeters are movable. Element 26: wherein the collecting further comprises moving the second set of the one or more surface quantum gravimeters after the integration time interval to respective new locations and collecting additional gravitational data using the second set of the one or more surface quantum gravimeters for the integration time interval at a next collection time interval. Element 27: further comprising a result transceiver, capable of communicating the analyzed gravitational parameters, the subterranean formation parameters, and interim outputs, to a user system, a data store, or a computing system. Element 28: wherein the computing system is one or more of a reservoir controller, a well site controller, or a non-quantum gravimeter. Element 29: wherein the set of quantum gravimeters collect the gravitational data at a collection time interval. Element 30: wherein the set of quantum gravimeters collect the gravitational data over an integration time interval. Element 31: further comprising a set of non-quantum gravimeters, capable to collect a second set of gravitational data, and wherein the data transceiver is further capable of receiving the second set of gravitational data from the set of non-quantum gravimeters. Element 32: wherein at least one of non-quantum gravimeters in the set of non-quantum gravimeters is calibrated utilizing the analyzed gravitational parameters or the subterranean formation parameters. Element 33: wherein the set of non-quantum gravimeters are MEMs gravimeters. Element 34: where at least one non-quantum gravimeter in the set of non-quantum gravimeters is located downhole a wellbore of the well sites. Element 35: wherein at least one quantum gravimeter in the set of quantum gravimeters is at a fixed surface location. Element 36: wherein at least one quantum gravimeter in the set of quantum gravimeters is mobile and is moved to a new location at a determined time. Element 37: wherein at least one quantum gravimeter in the

set of quantum gravimeters is located downhole a wellbore of the well sites. Element 38: wherein the one or more subterranean formation parameters include at least one of a depth parameter, a direction parameter, a fluid movement parameter, a gravimetric noise parameter, or a subterranean formation damage parameter. Element 39: further comprising a machine learning system, capable to communicate with the gravitational processor, and to process the gravitational data using one or more inversion algorithms. Element 40: wherein each gravimeter in the set of quantum gravimeters uses rubidium atoms. Element 41: wherein each gravimeter in the set of quantum gravimeters cools measuring atoms to less than 10 Kelvin.

Claims

1. A method, comprising: locating one or more surface quantum gravimeters at a surface location proximate a well site of a reservoir, where the well site is for moving fluids into or out of the reservoir and has one or more wellbores; collecting gravitational data at the one or more surface quantum gravimeters, including receiving input parameters at one or more of the one or more surface quantum gravimeters, wherein the input parameters specify one or more of an integration time interval or a collection time interval; and processing the gravitational data received from the one or more surface quantum gravimeters, generating analyzed gravitational parameters and subterranean formation parameters that indicate mass changes within a subterranean formation of the reservoir as a stage of a well site operation plan is executed.
2. The method as recited in claim 1, further comprising: modifying the well site operation plan using the analyzed gravitational parameters and the subterranean formation parameters.
3. The method as recited in claim 1, further comprising: communicating the analyzed gravitational parameters and the subterranean formation parameters to a well site controller, a reservoir controller, a computing system, a data center, or a cloud environment.
4. The method as recited in claim 1, wherein the well site is one of an injection well system and a first fluid is pumped downhole one or more of the one or more wellbores, the well site is a production well system and a second fluid is extracted from one or more of the one or more wellbores, or the well site is a hydraulic fracturing (HF) well system, and a HF fluid is pumped downhole one or more of the one or more wellbores.
5. The method as recited in claim 1, wherein the well site is an observation well system.
6. The method as recited in claim 1, wherein the collecting occurs at a collection time interval.
7. The method as recited in claim 1, wherein the gravitational data represents one of an absolute gravity reading, a relative gravity reading, or a combination of gravity readings.
8. The method as recited in claim 1, wherein one or more non-quantum gravimeters are proximate the well site, further comprising: calibrating the one or more non-quantum gravimeters using the gravitational data received from the one or more surface quantum gravimeters.
9. The method as recited in claim 1, wherein the processing determines one or more of a depth parameter of a subterranean fluid in the subterranean formation, a direction parameter to the subterranean fluid in the subterranean formation, a fluid movement parameter, a gravimetric noise parameter, or a subterranean formation damage parameter.
10. The method as recited in claim 9, wherein the gravimetric noise parameter is determined using a gravitational numerical model.
11. The method as recited in claim 1, wherein the integration time interval is in a range of 1 second to 1,000 seconds.
12. The method as recited in claim 1, wherein a first set of the one or more surface quantum gravimeters are at respective fixed locations and a second set of the one or more surface quantum gravimeters are movable, wherein the collecting further comprises: moving the second set of the one or more surface quantum gravimeters after the integration time interval to respective new locations and collecting additional gravitational data using the second set of the one or more

surface quantum gravimeters for the integration time interval at a next collection time interval.

13. The method as recited in claim 1, wherein one or more downhole gravimeters are located downhole one or more of the one or more wellbores, and the processing uses gravitational data from the one or more downhole gravimeters.

14. The method as recited in claim 13, wherein the one or more of the one or more downhole gravimeters are downhole quantum gravimeters.

15. The method as recited in claim 13, wherein the one or more of the one or more downhole gravimeters are non-quantum gravimeters.

16. The method as recited in claim 13, wherein the one or more downhole gravimeters are at respective fixed locations within the one or more wellbores.

17. The method as recited in claim 13, wherein the one or more downhole gravimeters have lesser sensitivity than the one or more surface quantum gravimeters.

18. A system, comprising: a set of quantum gravimeters positioned to collect gravitational data of a subterranean formation of a reservoir, wherein the reservoir has well sites for hydrocarbon extraction, the set of quantum gravimeters positioned to receive input parameters that specify one or more of an integration time interval or a collection time interval; a data transceiver, capable of receiving the gravitational data from the set of quantum gravimeters; and a gravitational processor, capable of communicating with the data transceiver and analyzing the gravitational data to generate analyzed gravitational parameters and one or more subterranean formation parameters.

19. The system as recited in claim 18, further comprising: a result transceiver, capable of communicating the analyzed gravitational parameters, the subterranean formation parameters, and interim outputs, to a user system, a data store, or a computing system.

20. The system as recited in claim 19, wherein the computing system is one or more of a reservoir controller, a well site controller, or a non-quantum gravimeter.

21. The system as recited in claim 18, wherein the set of quantum gravimeters receive the input parameters that specify the collection time interval.

22. The system as recited in claim 18, wherein the set of quantum gravimeters receive the input parameters that specify the integration time interval.

23. The system as recited in claim 18, further comprising: a set of non-quantum gravimeters, capable to collect a second set of gravitational data, and wherein the data transceiver is further capable of receiving the second set of gravitational data from the set of non-quantum gravimeters.

24. The system as recited in claim 23, wherein at least one of non-quantum gravimeters in the set of non-quantum gravimeters is calibrated utilizing the analyzed gravitational parameters or the subterranean formation parameters.

25. The system as recited in claim 23, wherein the set of non-quantum gravimeters are MEMS gravimeters.

26. The system as recited in claim 23, where at least one non-quantum gravimeter in the set of non-quantum gravimeters is located downhole a wellbore of the well sites.

27. The system as recited in claim 18, wherein at least one quantum gravimeter in the set of quantum gravimeters is at a fixed surface location.

28. The system as recited in claim 18, wherein at least one quantum gravimeter in the set of quantum gravimeters is mobile and is moved to a new location at a determined time.

29. The system as recited in claim 18, wherein at least one quantum gravimeter in the set of quantum gravimeters is located downhole a wellbore of the well sites.

30. The system as recited in claim 18, wherein the one or more subterranean formation parameters include at least one of a depth parameter, a direction parameter, a fluid movement parameter, a gravimetric noise parameter, or a subterranean formation damage parameter.

31. The system as recited in claim 18, further comprising: a machine learning system, capable to communicate with the gravitational processor, and to process the gravitational data using one or more inversion algorithms.

32. The system as recited in claim 18, wherein each gravimeter in the set of quantum gravimeters uses rubidium atoms.

33. The system as recited in claim 18, wherein each gravimeter in the set of quantum gravimeters cools measuring atoms to less than 10 Kelvin.

34. A computer program product having a series of operating instructions stored on a non-transitory computer-readable medium that directs a data processing apparatus when executed thereby to perform operations to determine subterranean formation parameters and analyzed gravitational parameters, the operations comprising: receiving collected gravitational data from one or more surface quantum gravimeters, wherein the one or more surface quantum gravimeters are located at a surface location proximate a well site of a reservoir, where the well site is for hydrocarbon production and has one or more wellbores, including receiving input parameters at one or more of the one or more surface quantum gravimeters, wherein the input parameters specify one or more of an integration time interval or a collection time interval; processing the gravitational data received from the one or more surface quantum gravimeters, generating the analyzed gravitational parameters and the subterranean formation parameters that indicate fluid movement in a subterranean formation of the reservoir as a stage of a well site operation plan is executed; and directing a modification the well site operation plan using the analyzed gravitational parameters and the subterranean formation parameters.

35. A method, comprising: locating one or more surface quantum gravimeters at a surface location proximate a well site of a reservoir, where the well site is for moving fluids into or out of the reservoir and has one or more wellbores; collecting gravitational data at the one or more surface quantum gravimeters; locating one or more non-quantum gravimeters proximate the well site; calibrating the one or more non-quantum gravimeters using the gravitational data received from the one or more surface quantum gravimeters; and processing the gravitational data received from the one or more surface quantum gravimeters, generating analyzed gravitational parameters and subterranean formation parameters that indicate mass changes within a subterranean formation of the reservoir as a stage of a well site operation plan is executed.

36. A method, comprising: locating one or more surface quantum gravimeters at a surface location proximate a well site of a reservoir, where the well site is for moving fluids into or out of the reservoir and has one or more wellbores; collecting gravitational data at the one or more surface quantum gravimeters; and processing the gravitational data received from the one or more surface quantum gravimeters, generating analyzed gravitational parameters and subterranean formation parameters that indicate mass changes within a subterranean formation of the reservoir as a stage of a well site operation plan is executed, wherein the processing determines one or more of a depth parameter of a subterranean fluid in the subterranean formation, a direction parameter to the subterranean fluid in the subterranean formation, a fluid movement parameter, a gravimetric noise parameter, or a subterranean formation damage parameter, and further wherein the gravimetric noise parameter is determined using a gravitational numerical model.

37. A system, comprising: a set of quantum gravimeters positioned to collect gravitational data of a subterranean formation of a reservoir, wherein the reservoir has well sites for hydrocarbon extraction; one or more non-quantum gravimeters positioned proximate the well site a data transceiver, capable of receiving the gravitational data from the set of quantum gravimeters; and a gravitational processor, capable of communicating with the data transceiver and analyzing the gravitational data to generate analyzed gravitational parameters and one or more subterranean formation parameters, the one or more non-quantum gravimeters configured to be calibrated using the gravitational data received from the set of quantum gravimeters.

38. A system, comprising: a set of quantum gravimeters positioned to collect gravitational data of a subterranean formation of a reservoir, wherein the reservoir has well sites for hydrocarbon extraction; a data transceiver, capable of receiving the gravitational data from the set of quantum gravimeters; and a gravitational processor, capable of communicating with the data transceiver and

analyzing the gravitational data to generate analyzed gravitational parameters and one or more subterranean formation parameters, including being capable of processing the gravitational data received from the one or more surface quantum gravimeters, generating analyzed gravitational parameters and subterranean formation parameters that indicate mass changes within a subterranean formation of the reservoir as a stage of a well site operation plan is executed, wherein the processing determines one or more of a depth parameter of a subterranean fluid in the subterranean formation, a direction parameter to the subterranean fluid in the subterranean formation, a fluid movement parameter, a gravimetric noise parameter, or a subterranean formation damage parameter, and further wherein the gravimetric noise parameter is determined using a gravitational numerical model.
